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C&S BIO M150 - Lab 3

Exercise 1 (50 pts):

Simulate Eqs.(1)-(2) beginning with three initial values in (m, p) space: (0, 0), (7, 25), and (1, 150). What do you expect to happen initially? Will both mRNA and protein increase in all cases? Where should the dynamics converge? Then, overlay the dynamics onto the velocity (vector) field for all three starting conditions, over a 24 hr simulation.

What do you expect to happen initially?:

Before simulating, I can predict the initial behavior by evaluating dm/dt and dp/dt at each IC:

- (0,0): dm/dt = +5, dp/dt = 0 → mRNA rises, protein stays flat meaning no mRNA has been made yet to translate
- (7,25): dm/dt = −2, dp/dt = +185 → mRNA decreases above m*, protein rises rapidly
- (1,150): dm/dt = +4, dp/dt = −120 → mRNA rises, protein decreases (too little mRNA to sustain p = 150)

Will both mRNA and protein increase in all cases?

No. Only IC (0,0) has both increasing initially. At (7,25), mRNA decreases. At (1,150), protein decreases. The sign of each rate depends on which side of the corresponding nullcline the IC lies on.

Where should the dynamics converge?

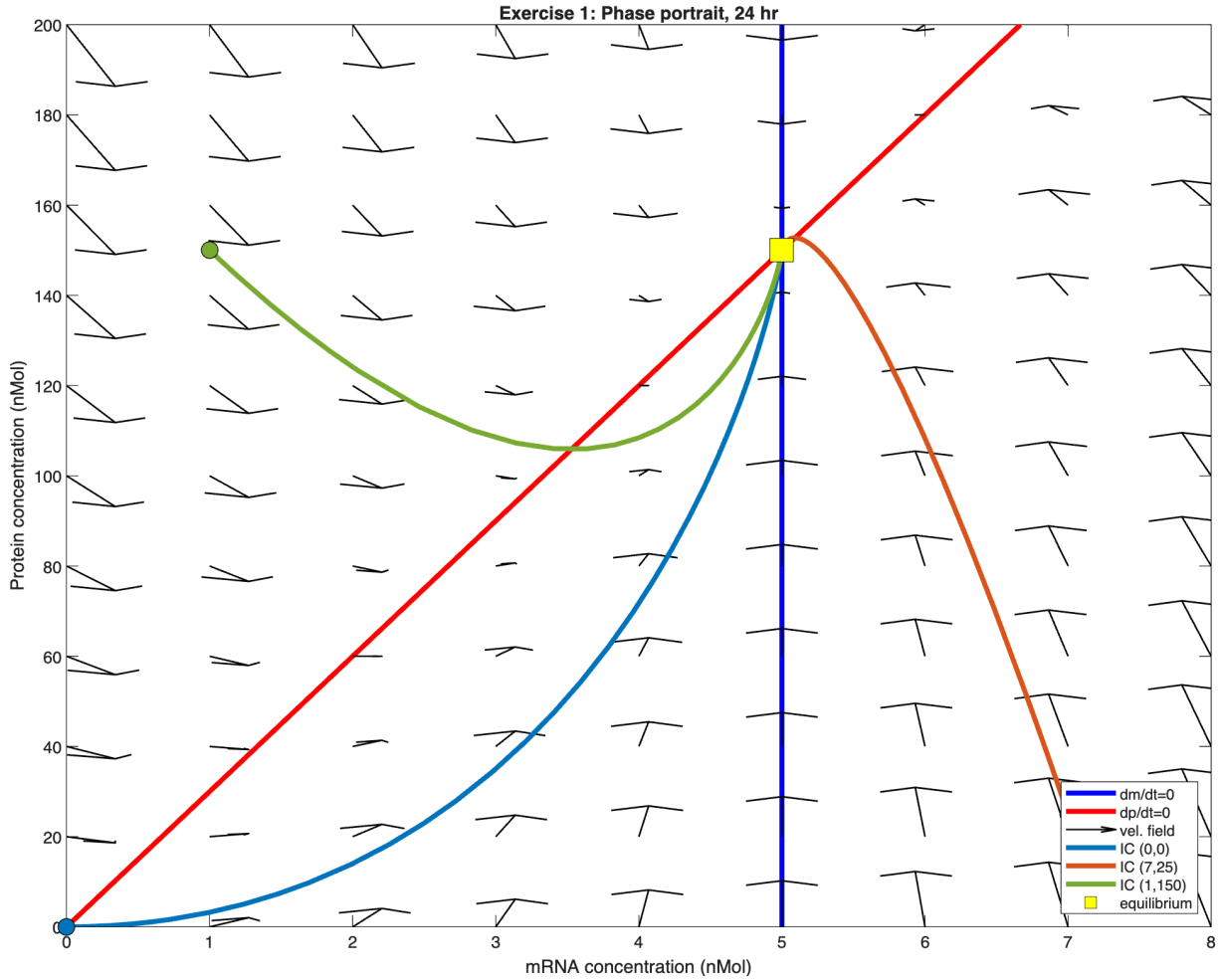
They should converge at the intersection of the nullclines: m* = a/b = 5nMol, p* = ra/b^2 = 150 nMol. The simulation from the code confirms that all three ICs reach the 5, 150 by t = 24 hr as shown overlaid on on the vector field below.

```
% copied and pasted given code from the lab document
pars.a = 5;
pars.b = 1;
pars.r = 30;

t0 = 0;  tf = 24;
ICs    = [0 0; 7 25; 1 150];
colors = [0 0.45 0.74; 0.85 0.33 0.10; 0.47 0.67 0.19];

mnullcline = pars.a/pars.b;
pnullcline = @(p) pars.b.*p./pars.r;
pvec = 0:20:200;
mvec = 0:1:8;

dmdt = @(m)   pars.a - pars.b.*m;
dpdt = @(m,p) pars.r.*m - pars.b.*p;
[MM,PP] = meshgrid(mvec,pvec);

% plotting for exercise 1
figure(1); clf;
set(gcf, 'Position', [100, 100, 1200, 900]);
```

```
% vertical mRNA nullcline (m = a/b)
plot(mnullcline*ones(size(pvec)), pvec, 'b', 'LineWidth',3); hold on;

 % diagonal protein nullcline (m = bp/r)
plot(pnullcline(pvec), pvec, 'r', 'LineWidth',3);

% velocity field: arrows showing (dm/dt, dp/dt) at each point on the grid
quiver(MM,PP,dmdt(MM),dpdt(MM,PP),'k','MaxHeadSize',0.015,'LineWidth',1);

% plotting and running ODE simulation for each pair of initial values
for k = 1:size(ICs,1)
    x0 = ICs(k,:);
    [t,x] = ode45(@(t,x) proteinproduction(t,x,pars), [t0 tf], x0);
    plot(x(:,1), x(:,2), '-',  'Color', colors(k,:), 'LineWidth', 3);
    plot(x0(1),  x0(2),  'o',  'MarkerFaceColor', colors(k,:), ...
         'MarkerEdgeColor','k','MarkerSize',10);
    % printing out final values corresponding to the initial pair of state
    % values
    fprintf('IC=(%g,%g): initial dm/dt=%+g, dp/dt=%+g | final
(m,p)=(%.4f,%.4f)\n', ...
            x0(1), x0(2), dmdt(x0(1)), dpdt(x0(1),x0(2)), x(end,1),
x(end,2));
end
```

```
IC=(0,0): initial dm/dt=+5, dp/dt=+0 | final (m,p)=(5.0000,150.0000)
IC=(7,25): initial dm/dt=-2, dp/dt=+185 | final (m,p)=(5.0000,150.0000)
IC=(1,150): initial dm/dt=+4, dp/dt=-120 | final (m,p)=(5.0000,150.0000)
```

```
% marking the equilibrium point of intersection with a yellow square
plot(pars.a/pars.b, pars.r*pars.a/pars.b^2, 's', ...
     'MarkerFaceColor','y','MarkerEdgeColor','k','MarkerSize',18);
hold off;
% setting ranges for the plot
xlim([min(mvec) max(mvec)]);
ylim([min(pvec) max(pvec)]);
xlabel('mRNA concentration (nMol)','FontSize',18);
ylabel('Protein concentration (nMol)','FontSize',18);
legend({'dm/dt=0','dp/dt=0','vel. field', ...
        'IC (0,0)','','IC (7,25)','','IC (1,150)','','equilibrium'}, ...
       'Location','southeast');
set(gca,'FontSize',10);
title('Exercise 1: Phase portrait, 24 hr');
```



Exercise 2 (50 pts):

Can you characterize the shape of the return of the concentration of proteins to equilibrium with time (0-5 hours for this problem)? Can you predict what it should be from the linearized system of equations and verify the quantitative rate of return? Verify this by estimating the fixed point and rate of return for the following parameter sets: (a = 5, b = 1, r = 30), (a = 5, b = 0.5, r = 30) and (a = 10, b = 1, r = 30).

Can you characterize the shape of the return of the concentration of proteins to equilibrium with time (0-5 hours for this problem)?

To characterize how protein returns to equilibrium in terms of their shape, I linearize the system around the fixed point using small deviations $u = m - m^*$ and $v = p - p^*$:

$du/dt = -bu$

$dv/dt = ru - bv$

The Jacobian matrix from these state variables is $J = [-b\ 0;\ r\ -b]$. Its eigenvalues come from $\det(J - \lambda I) = (\lambda + b)^2 = 0$, giving a repeated eigenvalue $\lambda = -b$. Because both eigenvalues are negative, the equilibrium is locally stable.

By solving, the linearized system, it gives the protein dynamics near equilibrium:

$p(t) - p^* = [(p_0 - p^*) + r(m_0 - m^*)t] \cdot \exp(-bt)$

The equation has two parts: an exponential decay (exp(-bt)) that pulls the protein back to p* which is then multiplied by a linear term in t from the mRNA being included in the protein equation. This is why the curves in the plots below are S-shaped instead of like an exponential function. The linear factor helps delay the decay at early time before the exponential part takes over.

Can you predict what it should be from the linearized system of equations and verify the quantitative rate of return?

Yes, I can. The linearized system has Jacobian $J = [-b\ 0;\ r\ -b]$ with repeated eigenvalue $\lambda = -b$, predicting a return rate of $\exp(-bt)$. This can be verified quantitatively by simulating small perturbations from the fixed point and comparing the decay of $|p(t) - p^*|$ against the predicted rate $\lambda = -b$.

The fitted rates (–0.85, –0.39, –0.85) approximately match the theoretical $\lambda = -b$ (–1, –0.5, –1).

From my code below to verify the quantitative rate of return:

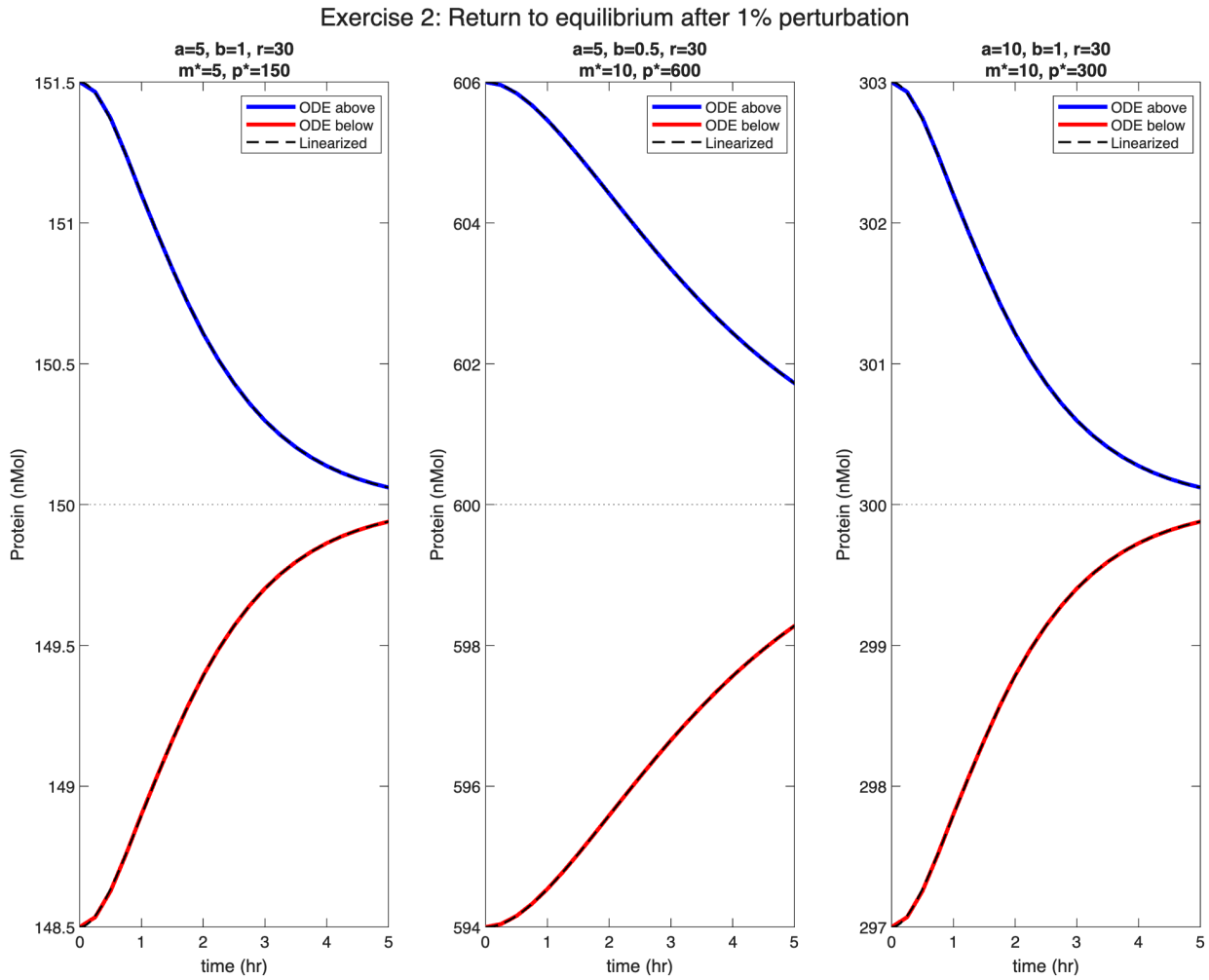
```
Set 1 (a=5, b=1, r=30): m*=5, p*=150 | theoretical rate = -1, fitted = -0.8486

Set 2 (a=5, b=0.5, r=30): m*=10, p*=600 | theoretical rate = -0.5, fitted =
-0.3879

Set 3 (a=10, b=1, r=30): m*=10, p*=300 | theoretical rate = -1, fitted = -0.8486
```

```
clear pars; figure(2); clf;
set(gcf, 'Position', [100, 100, 1200, 900]);

% values to be tested
sets = [ 5   1.0  30;
         5   0.5  30;
        10   1.0  30 ];

% testing each trio of values and plotting them as well
for k = 1:size(sets,1)
    pars.a = sets(k,1);
    pars.b = sets(k,2);
    pars.r = sets(k,3);

    % fixed point
    mstar = pars.a/pars.b;
    pstar = pars.r*pars.a/pars.b^2;

    % verify fzero finds the same thing
    allnull   = @(p) pars.r*pars.a/pars.b - pars.b*p;
    pstar_num = fzero(allnull, 100);

    % 1% perturbations above and below the fixed test point to test
    % stability
    x0_above = [mstar*1.01  pstar*1.01];
    x0_below = [mstar*0.99  pstar*0.99];

    % simulating ODE from each pertubed IC (1% above and 1% below) up to t
= 10 hr
```

```
    [ta,xa] = ode45(@(t,x) proteinproduction(t,x,pars), [0 10], x0_above);
    [tb,xb] = ode45(@(t,x) proteinproduction(t,x,pars), [0 10], x0_below);

    % Linearized prediction for protein production:
    % p(t) - p* = [(p0-p*) + r(m0-m*)t] * exp(-bt)
    p_lin = @(t,m0,p0) pstar + ((p0-pstar) + pars.r*(m0-mstar).*t).*exp(-
pars.b.*t);

    % plotting the ODE solutions in solid lines
    % plotting the linear solutions (predictions) in dashed lines
    subplot(1,3,k);
    % blue line is 1% pertubed above
    plot(ta, xa(:,2), 'b', 'LineWidth',2.5); hold on;
    % red line is 1% pertubed below
    plot(tb, xb(:,2), 'r', 'LineWidth',2.5);
    tfine = linspace(0,10,500);
    % linear solutions (predictions)
    plot(tfine, p_lin(tfine,x0_above(1),x0_above(2)),
'k--','LineWidth',1.3);
    plot(tfine, p_lin(tfine,x0_below(1),x0_below(2)),
'k--','LineWidth',1.3);
    % dotted reference line at p*
    yline(pstar,':');
    xlim([0 5]);

    % labeling
    xlabel('time (hr)','FontSize',14);
    ylabel('Protein (nMol)','FontSize',14);
    title(sprintf('a=%g, b=%g, r=%g\nm*=%g, p*=%g', ...
                  pars.a, pars.b, pars.r, mstar, pstar));
    legend({'ODE above','ODE below','Linearized'},'Location','best');
    set(gca,'FontSize',10);

    dev  = abs(xa(:,2) - pstar);
    keep = ta > 2/pars.b & dev > 1e-8;     % window where exp dominates
    slope = polyfit(ta(keep), log(dev(keep)), 1);
    fprintf(['Set %d (a=%g, b=%g, r=%g): m*=%g, p*=%g | ' ...
             'theoretical rate = %+g, fitted = %+.4f\n'], ...
            k, pars.a, pars.b, pars.r, mstar, pstar, -pars.b, slope(1));
end
```

```
Set 1 (a=5, b=1, r=30): m*=5, p*=150 | theoretical rate = -1, fitted = -0.8486
Set 2 (a=5, b=0.5, r=30): m*=10, p*=600 | theoretical rate = -0.5, fitted = -0.3879
Set 3 (a=10, b=1, r=30): m*=10, p*=300 | theoretical rate = -1, fitted = -0.8486
```

```
sgtitle('Exercise 2: Return to equilibrium after 1% perturbation');
```



Note: The linearized curve (dashed line) overlays the ODE graphs, confirming the previous prediction.

```
function dxdt = proteinproduction(t,x,pars)
% proteinproduction  RHS of mRNA-protein ODE system
%   dm/dt = a - b*m
%   dp/dt = r*m - b*p
    m = x(1);
    p = x(2);
    dmdt = pars.a - pars.b*m;
    dpdt = pars.r*m - pars.b*p;
    dxdt = [dmdt; dpdt];
end
```